Lab 02 – Data Preparation

Analytics and Knowledge Discovery, WS 2025 – summary of exercises DP1–DP6 and the key points to learn.

DP1 – Removing rows with missing data

Dataset: gender, age, preference score (0–10); 242 of 1000 preference values are missing. Goal: the mean preference.

- Missing values occur **only for people older than 40** (about 50 % of them), never for younger people.
- Preference increases strongly with age (correlation 0.92).
- Deleting incomplete rows removes mainly old people with high scores, so the mean drops: **4.53 after deletion vs. about 4.99** after correcting for age.



Key point: Deleting rows (complete-case analysis) is only safe when data is **Missing Completely At Random (MCAR)**. If missingness depends on another variable (MAR) or on the value itself (MNAR), the remaining data is biased. Always check *who* is missing before you delete anything.

DP2 – Imputation methods for time series

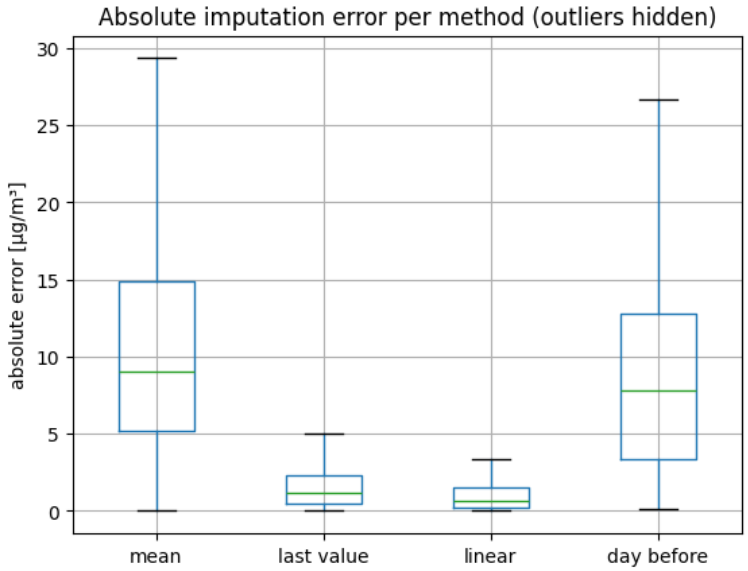
PM10 in Salzburg-Lehen, one value every 30 min (48 per day), 300 single values missing. Imputation means filling the gaps with a good estimate. The four methods, in pandas:

Method	Idea	pandas code	MAE
(a) mean	mean of the whole series	s.fillna(s.mean())	11.82
(b) last value	value just before (extrapolation)	s.ffill()	2.11
(c) linear interpolation	line between before and after	s.interpolate()	1.36
(d) day before	value 24 h earlier	s.fillna(s.shift(48))	9.85

Key point: Time series have **autocorrelation**: neighbouring values are similar. Methods that use the close neighbours (last value, interpolation) work much better than the global mean, which ignores time completely.

DP3 – Evaluating imputation with EDA

- **(a) Best method: linear interpolation** (MAE 1.36, RMSE 2.48), as expected, because it uses information from both sides of the gap.
- **(b) Is the error high?** No. Compared with a reference (mean PM10 about 21, std about 17) an MAE of 1.4 is small. Only 1.7 % of values are imputed and the mean/std of the series stay the same, so later analyses should not be harmed.
- **(c) Distribution:** errors are centred at 0 (no bias) with long tails. From the ECDF: P(relative error > 5 %) is about 42 %, P(> 20 %) about 8 %. Even these cases are small in absolute terms (median about 1.75 µg/m³).
- **(d)** Plotted on the full series, the imputed points lie on the curve and create no artificial peaks.

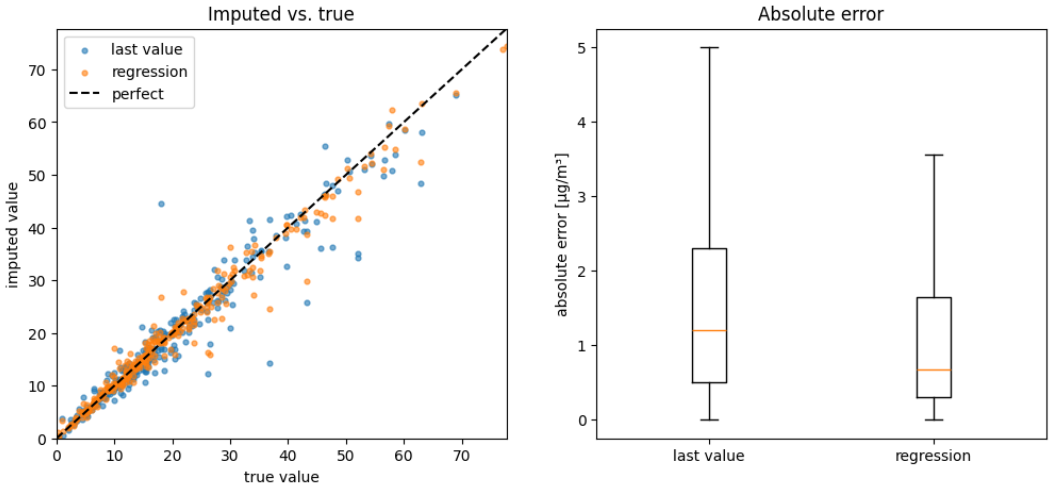


Key point: For evaluation you need three things: a **suitable graphic** (boxplot to compare methods, histogram + ECDF for distributions, time plot for context), a **suitable error measure** (MAE, RMSE, relative error) and a **reference** to judge whether an error is big.

DP4 – Imputation with linear regression

The data file only contains Lehen, so the 'other variables' are time-shifted copies of the series: previous / next value, ±1 h and ±24 h.

- Highest correlation with PM10: **prev and next (0.95)**; the 24 h shifts reach only 0.56.
- Model: **PM10 = 0.14 + 0.43·prev + 0.56·next**. The regression learned a weighted interpolation by itself.
- MAE: regression 1.42 vs. last value 2.11, so the regression is clearly better than the last value and close to interpolation.

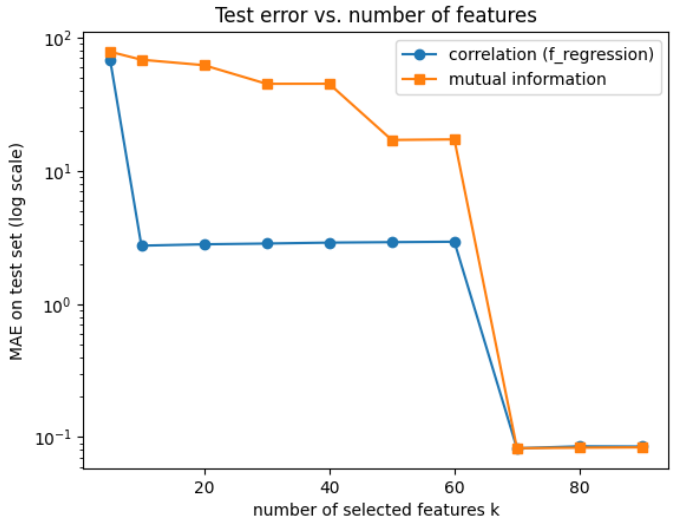


Key point: Regression imputation: pick predictors with **high correlation** to the missing variable, train on complete rows, predict the missing ones. A scatter plot of imputed vs. true values (diagonal = perfect) is a good way to compare methods.

DP5 – Feature selection: correlation vs. mutual information

Synthetic data: 100 features, only 10 informative. `SelectKBest` keeps the k best features; then a linear regression is trained and the test MAE is measured.

- **Correlation (f_regression)** scores the linear relationship between each feature and y.
- **Mutual information** measures any (also non-linear) dependence, but it has to be estimated and is noisy.
- Correlation is clearly better for k = 10–60. Both reach the minimum MAE of about 0.08 from **k = 70**.
- **Choose k of about 70**, the smallest k with minimal error. For linear data, correlation is the better method.

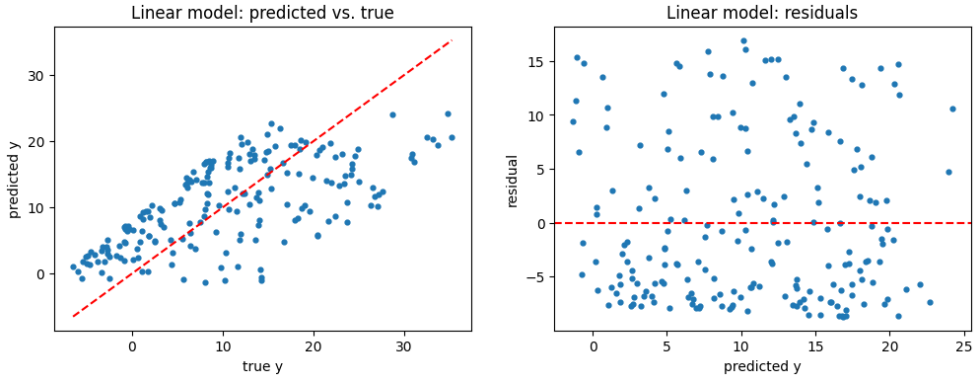


Key point: Fit the selector **only on training data** to avoid data leakage. k is a hyperparameter: try several values and pick the smallest k that reaches the lowest test error. Match the selection method to the model (linear model, so use correlation).

DP6 – Feature generation for linear regression

- Purely linear model $y = b0 + b1 \cdot x1 + b2 \cdot x2$: **$R^2 = 0.42$**. Predicted vs. true is far from the diagonal and the residuals show a pattern.

- With polynomial features ($x1^2$, $x1 \cdot x2$, $x2^2$): **$R^2 = 1.0$**. The coefficients reveal the true formula **$y = 3 + 2 \cdot x1 - x2 + x2^2$**.
- So yes, the polynomial model is much better.



Key point: A linear model is linear in its **coefficients**, not necessarily in its inputs. Adding generated features (squares, products, logs) lets it fit curves. **Residual plots** show when something is missing. Too many terms cause overfitting, so validate on test data.

Big picture: what to remember

Topic	Remember
Missing data	Check whether missingness is random before deleting rows.
Imputation	Use the structure of the data (time neighbours, correlated variables).
Evaluation	Graphic + error measure + reference.
Feature selection	Fewer, informative features. Choose k by test error. No leakage.
Feature generation	New features (x^2, $x \cdot y$) make linear models more powerful.